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Notes:
https://docs.google.com/document/d/11OZKpkt0Bw6rotXu3rIeST_58ZmPg77WJUlH71LUkJI/edit?tab=t.0
Assets from PMM: https://red.ht/virt-asset-deck
Architecture guide: https://access.redhat.com/articles/7119411

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OpenShift Virtualization DR -- State of the Union

On OpenShift

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Sr. Principal Software Engineer

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Agenda

- DR options
- Backup and restore -- Continuous Restore
- Volume Replication
    - Active/Active (vendors)
    - Active/Passive (vendors)
        - DR orchestration
- Red Hat Work in Progress

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Agenda

- OpenShift Disaster Recovery Technology Landscape
  - DR Overview
  - Continuous Restore
  - Active/Active or Metro DR
  - Active/Passive or Regional DR
- Red Hat Work in Progress
  - Contacts
  - Workstreams

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Agenda

OpenShift Disaster Recovery Technology Landscape

OpenShift Disaster Recovery Technology Landscape
- DR Overview
- Continuous Restore
- Active/Active or Metro DR
- Active/Passive or Regional DR

Red Hat Work in Progress
- Contacts
- Workstreams

Red Hat Work in Progress

Contacts

Workstreams

High-level DR Approaches

Failure Domain is either a data center or a cloud region

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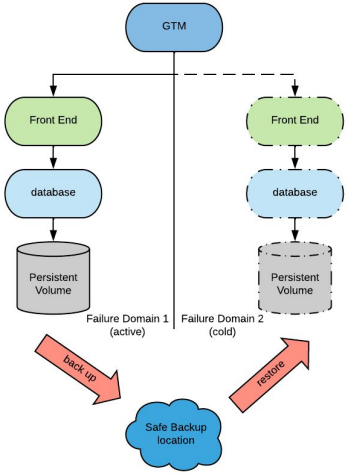
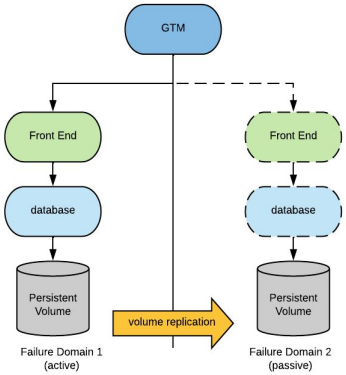
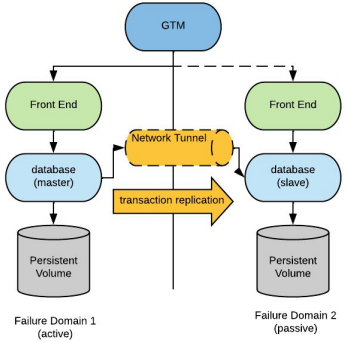
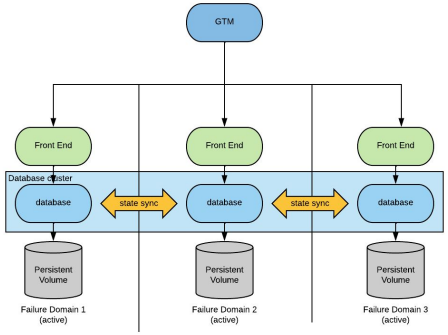
Active/Passive

Active/Active









Backup / Restore

RPO ~= frequency of backup
RTO ~= hours

Volume Replication

Asynchronous:

- RPO ~= unbounded (typically ms)
- RTO ~= typically minutes *

Synchronous:

- RPO = 0
- RTO ~= typically minutes*

Transaction Replication

RPO = 0
RTO ~= minutes

Distributed Stateful Workloads

Strong Consistency:

- RPO = 0
- RTO ~= seconds

Eventual Consistency (**)

- RPO ~= 0
- RTO ~= seconds

(*) a database can take an unbounded amount of time to recover from a crash consistent volume

(**) eventual consistency does not imply eventual correctness

High-level DR Approaches

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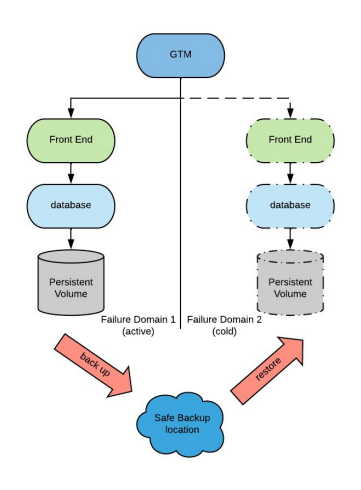
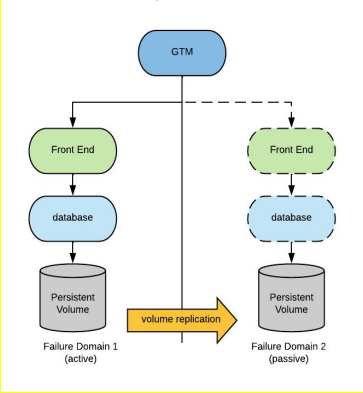
Failure Domain is either a data center or a cloud region

Applicable to VMs

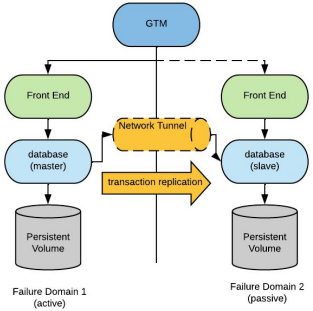
Active/Passive

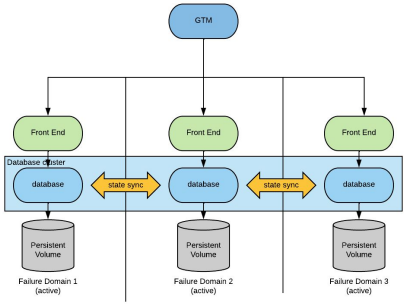




NO CSI API

Active/Active





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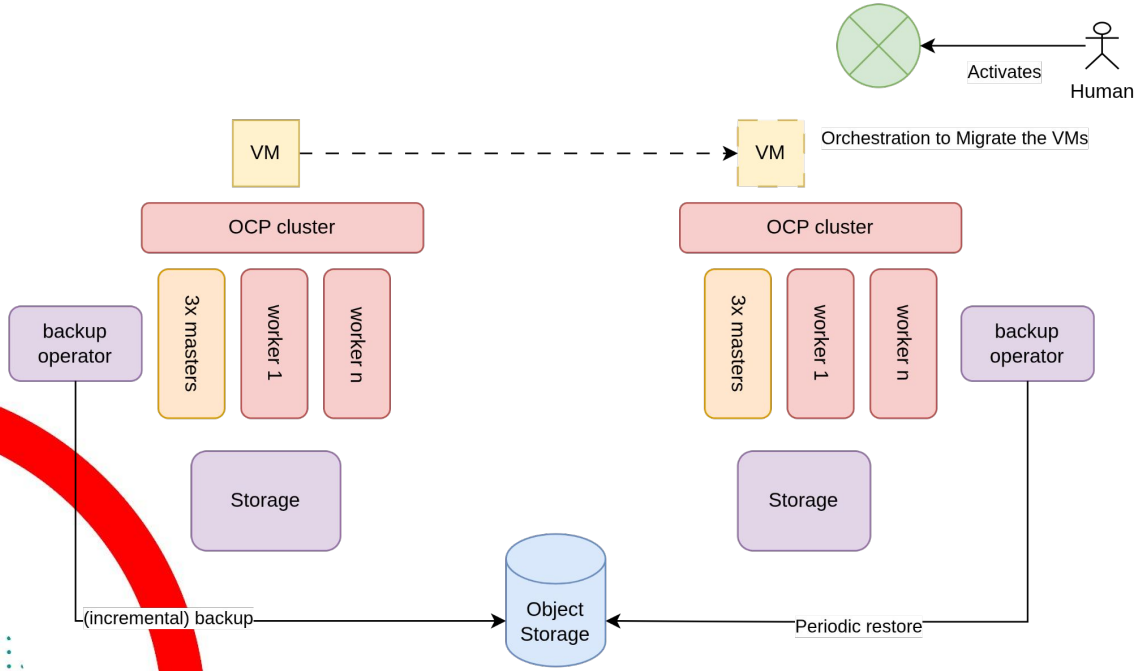
Continuous restore as a DR strategy



Considerations

- If customer really only has two failure domain, where is the object storage located?
- What network is used for the continuous backup and restore?
- Is Changed Block Tracking (CBT) available?
- What is the RPO? (I have not seen anything below 15 minutes)

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Backup and Restore Vendors

Red Hat is not a backup and restore company, do not use OADP by itself. Help the customer keep using their current product they use (prefered option), or select a product that is certified for OCP.

VERITAS™

Commvault®

COHESITY

rubrik

TRILIO

kasten by Veeam

Symmetric replication

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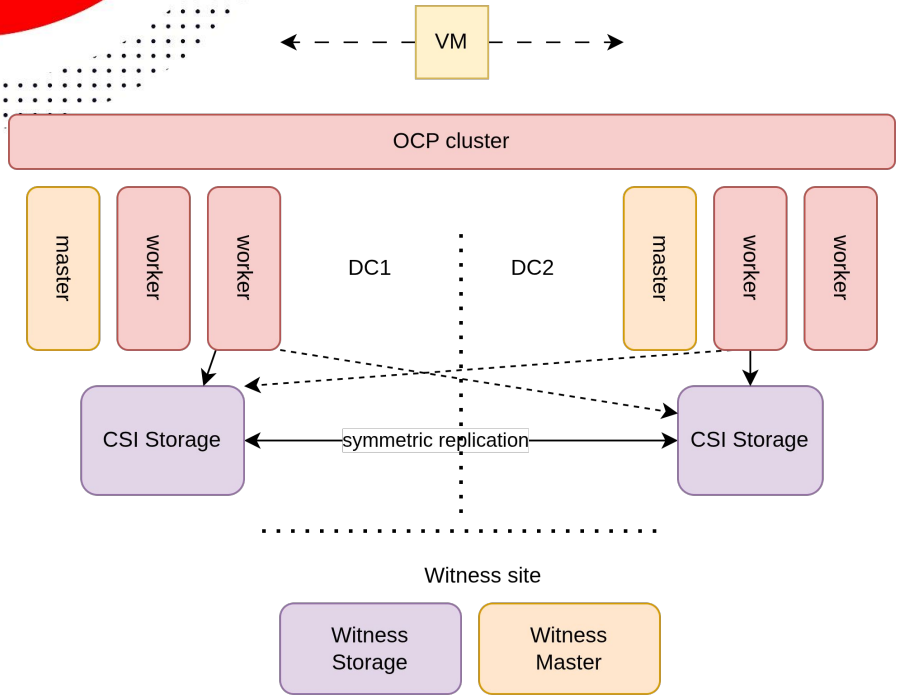
A.k.a MetroDR -- Active/Active -- Latency <5-10ms between DC1 and DC2



- OCP and/or Storage are stretched across DC1 and DC2
- A witness site is required for both OCP and storage
- Very little latency between DC1 and D2, more latency is usually tolerated between the witness site and the main DCs
- In case of a disaster VMs will restart in the healthy DC.
- In case the storage fails, the CSI driver should implement the failover of the multipath connection.
- VMs must be able to move without changing their IPs
- Labels and affinity rules might be used to have VMs prefer one DC or the other.
- CAUTION: a DR event may cause a VM restart storm, verify that storage and OCP control plane do not get overwhelmed.
- CAUTION: no restart order can be enforced (typically needed for DR plans), pod priority or node labels can be used partially control this behavior.

Note: some schools of thought do not consider this DR because the OCP control plane is a failure domain and if it breaks you lose the cluster
Note: Witness Master here means a normal master deployed in the witness site
Note: ensure to have at least 50% of available capacity in either side to support the VMs swinging during a DR event.

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Major Vendors with Support for Symmetric Replication

CSM extensions
+ repctl

HRPC extensions

ActiveCluster + CSI
Configuration

APP or CPP +
CSI configuration

MetroCluster +
CSI configuration

Hyperswap + CSI
configuration

Mono-directional replication

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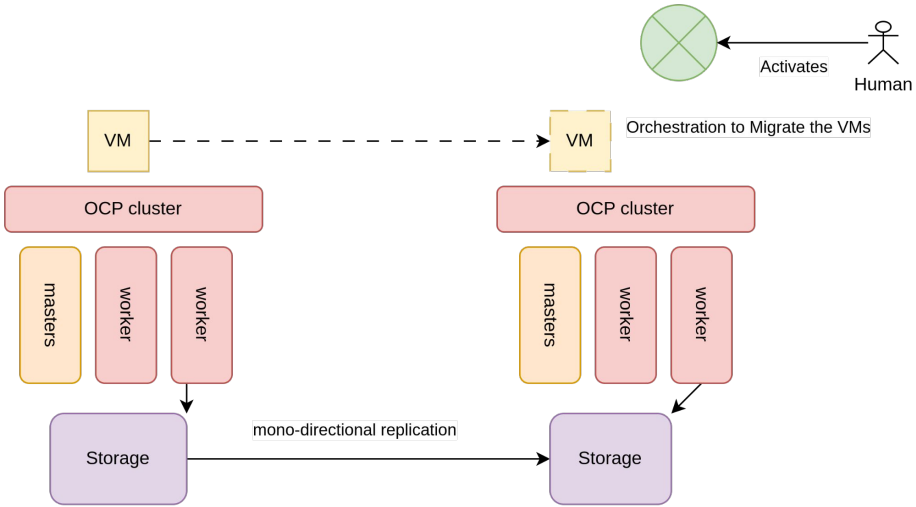
A.k.a Regional DR -- Active/Passive -- Latency > 10ms between DC1 and DC2



- VM Volumes are replicated to the other site (either sync or async replication)
- DR must be triggered externally, typically by a human.
- The DR process requires the following components:

Component	Description
Volume Group support	Ability to consistently replicate a group of volumes
Volume Replication Status management	Manage the direction of the replication and whether replication sia active or not.
VM Management	Ability to start/stop VMs in the corect DC, attaching them to the same Volumes
Recovery orchestration	Process that restarts all the VMs in the healthy site in case of a disaster, throttling the VM restarts and managing dependencies,

Conceptual state diagram of a DR orchestration

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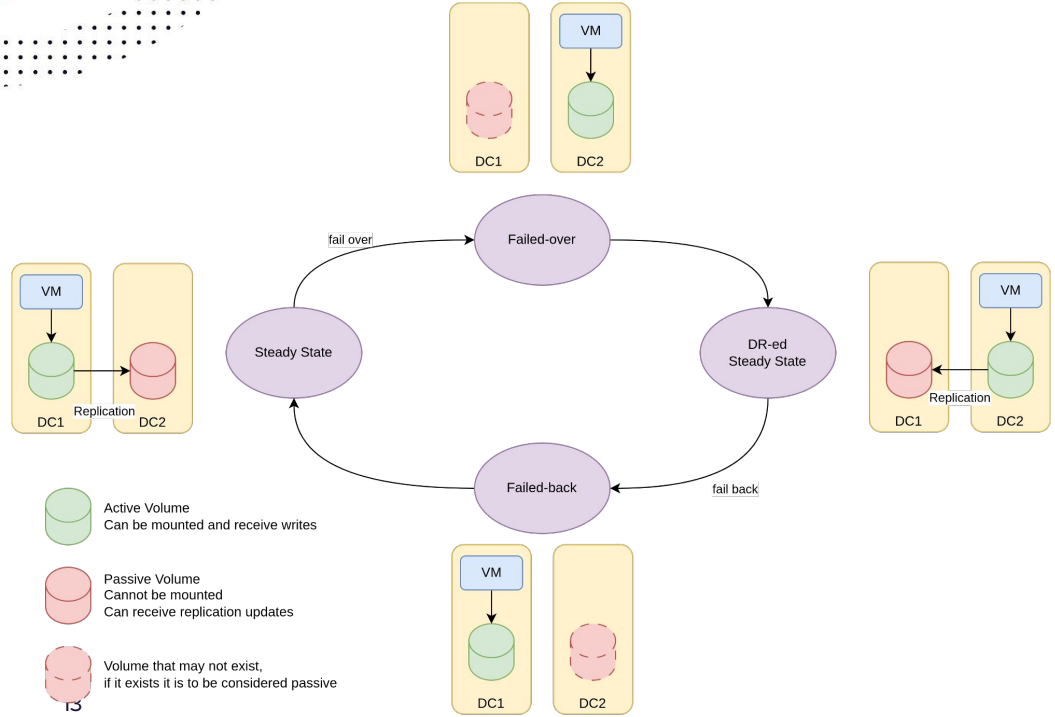

State transitions should be applicable to individual VMs or groups of VMs in a single transaction. This would allow to support the use cases of:

1. Individual application failover
2. Failover of all the application in a tier
3. Manage throttling and precedences

Vendors expose the capabilities to create this orchestration in very different way. There is not today a storage agnostic way of writing this orchestration.
There are many ways of writing this orchestration.
Ansible and gitops should be our go-tos

Major Vendors with support for unidirectional replication

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DELL

CSM extensions
+ repctl

HITACHI

HRPC extension

IBM
IBM FlashSystems™

CSI extension
(same as ODF)

HPE

CSI extension

NetApp™

CSI extension

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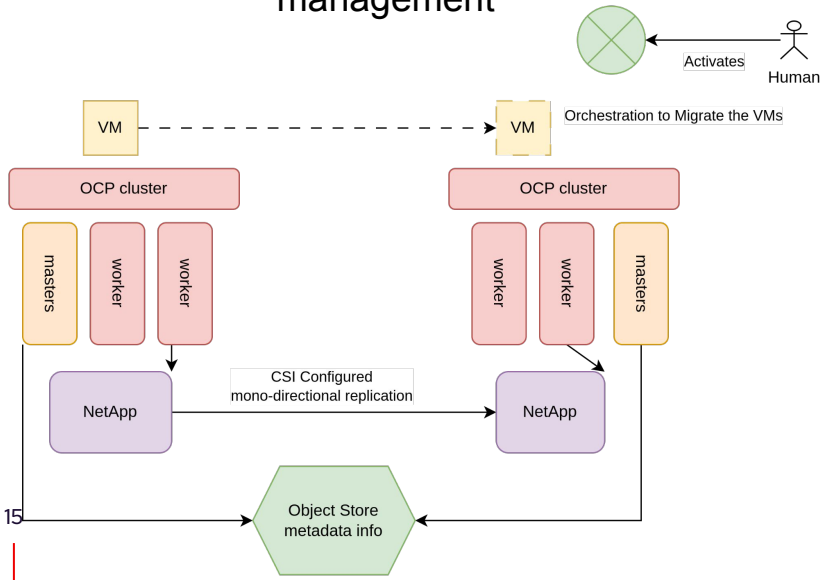
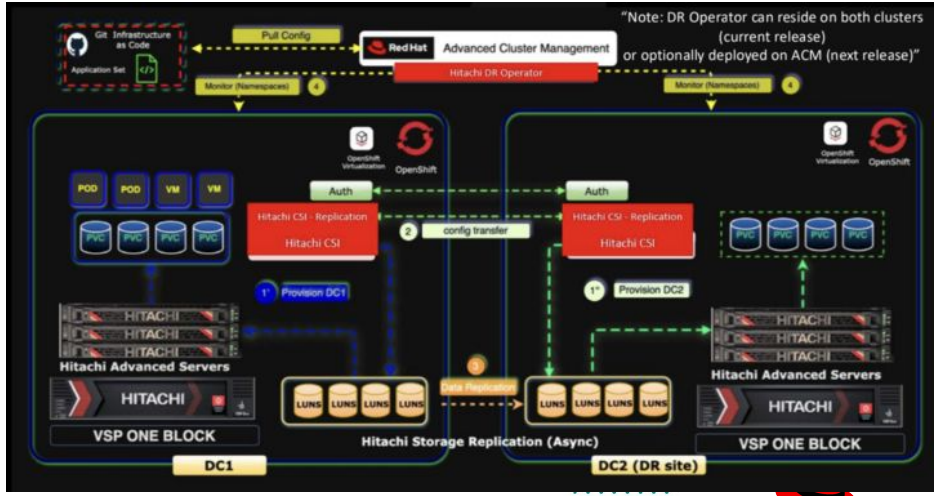
Notable Mentions

NetApp - metadata management



Hitachi - DR orchestrator



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What about SDS and VMFS-like storage options?

SDS

- ODF
- Portworx
- IBM Fusion

VMFS-like

- Arctera Infoscale
- Portworx data store
- IBM Fusion for SAN

Common considerations

- Good monodirectional volume replication story
- No or poor bidirectional replication, this is by design because it’s just simpler to extend the storage cluster
- Replication occurs over the network, handled by the openshift nodes. Caution must be taken to keep the network traffic isolated

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What is Red Hat Working on?

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Team from Ecosystems Engineering

Engineering, Product Management, and QE Resources engaged

Key Contacts:
Product Management - Peter Lauterbach
Engineering Manager - Andrew Beekhof
Architect - Martin Jackson

Work in Process
Subject to Change

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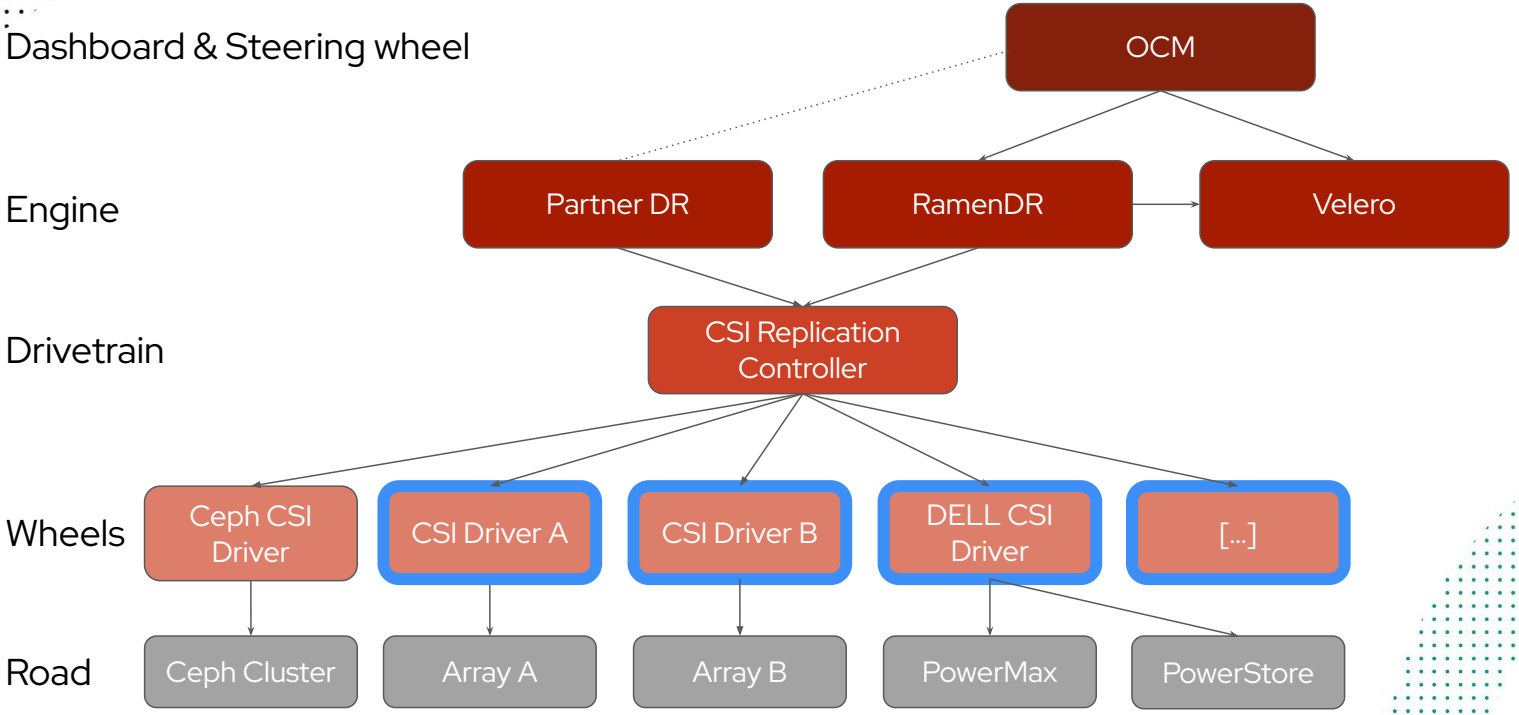
Preferred Path – Partner Acceleration

Work with storage partners to encourage and accelerate adoption of the CSI Replication Addon



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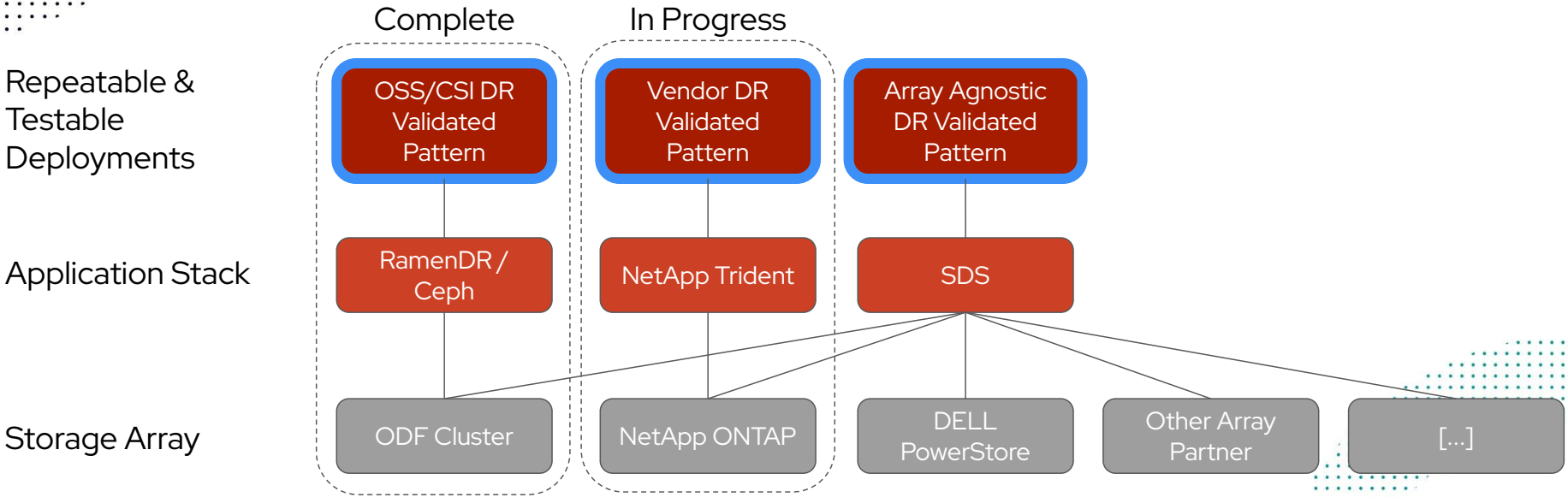
Done/In Process - Assemble

No new functionality



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OpenShift Virtualization DR in Summary

- Many flavors of disaster recovery to meet business requirements
- Solutions available from our valuable Data Protection partners
- All major storage vendors can support DR with OpenShift
- Some exciting future capabilities through collaborative engineering with our ecosystem partners

http://red.ht/virt-asset-deck

Useful assets for your OpenShift Virtualization conversations http://red.ht/virt-asset-deck

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